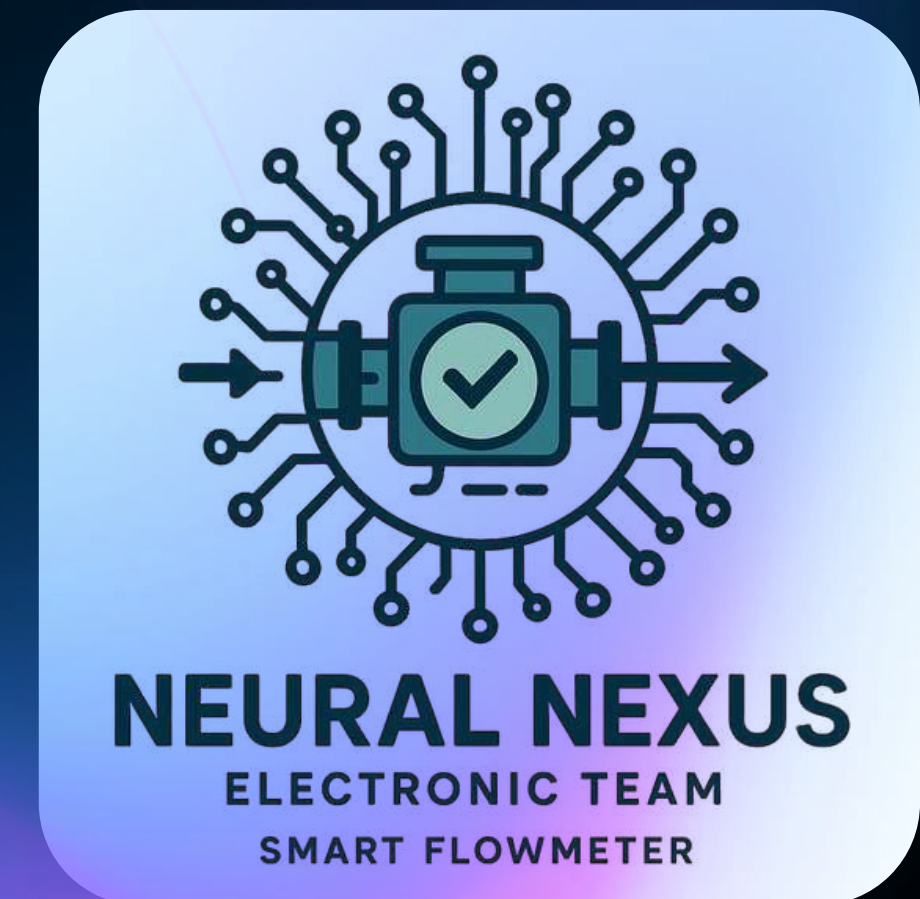


Smart Flow Meter

An IoT-integrated smart flowmeter with real-time monitoring of fluid flow suited for industrial and domestic applications. Equipped with advanced sensors and wireless connectivity the flowmeter continuously collects and transmits flow data to a cloud-based platform.



Team Members



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What differentiates us



01

Real time data

Provides real time data of flow rate and temperature of the fluid that flows through the flowmeter.



02

Analytical data

Records the flow rate and temperatures to the cloud server, which can be used to predict the future flow by analyzing the data



03

Special App features

Dynamic Control over the flowmeter. Multiple flowmeter readings are available which can be used for cross connected calculations

Market Analysis

Growing demand for flowmeters

The flow meter market size is projected to increase from USD 9.1 billion in 2024 to USD 12.6 billion by 2029, at a CAGR of 6.7%. The global flowmeters market growth is primarily driven by the demand from oil & gas and wastewater treatment industries

Special features

There is a market gap for a smart flowmeter which is portable , which can be integrated to any system to monitor flow rate , volume passed and current temperature through an app

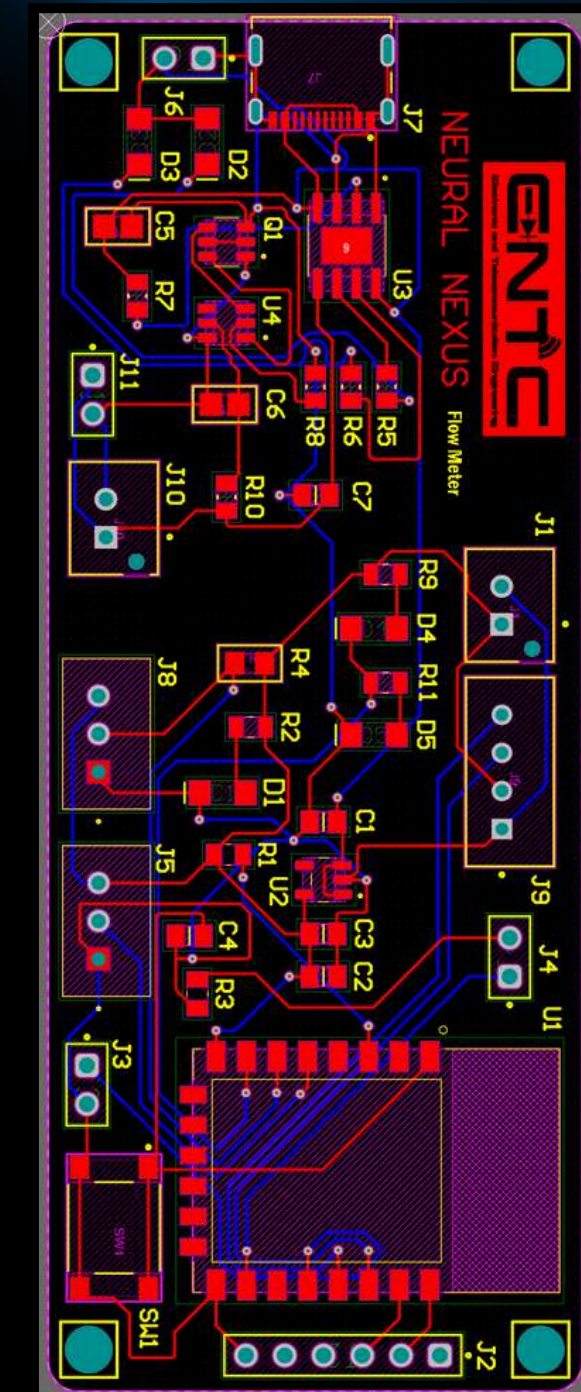
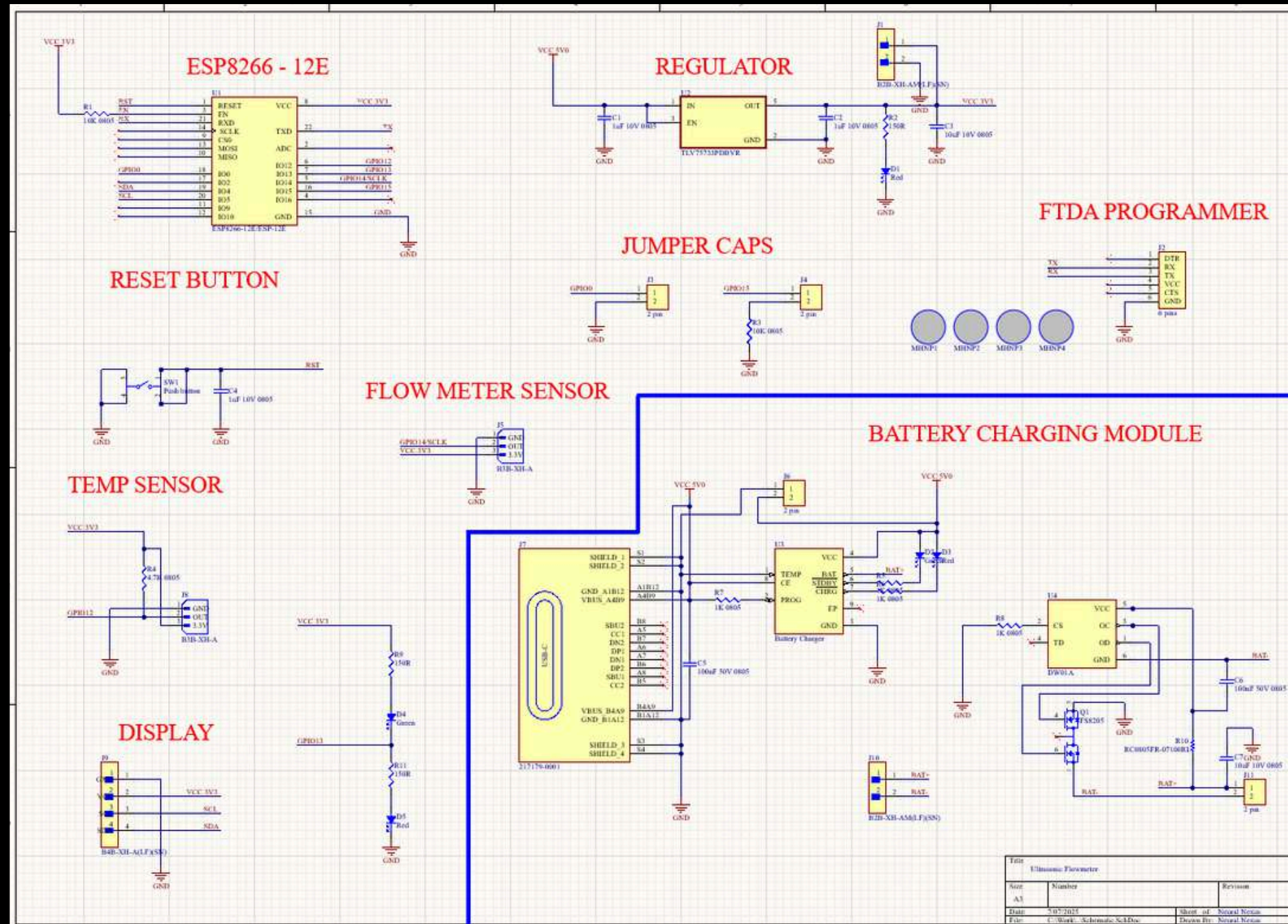
Wide Audience

Industries such as
Oil & Gas, Water & Wastewater, Chemical, Food & Beverage, Power Generation, Pharmaceuticals
Domestic Appliances

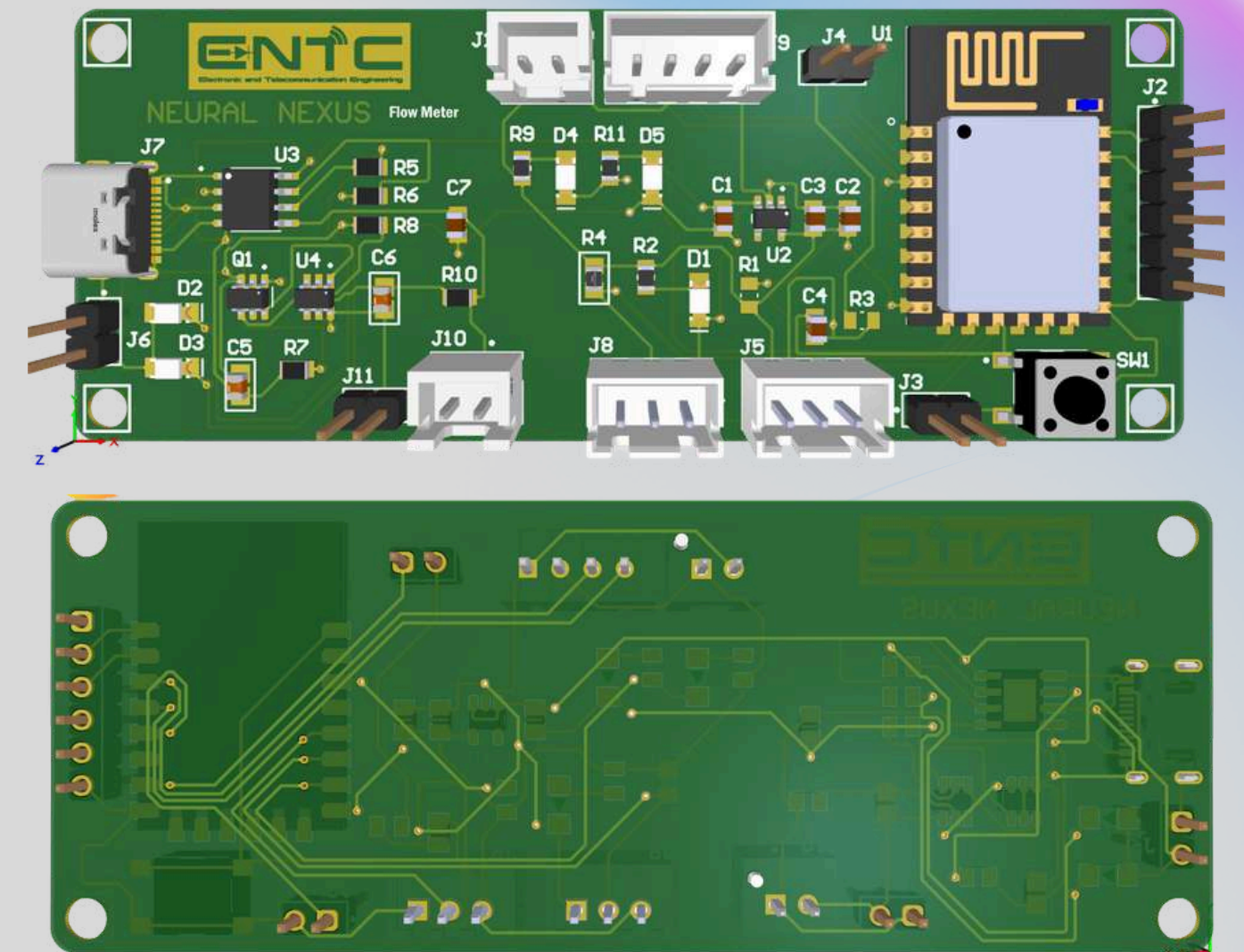
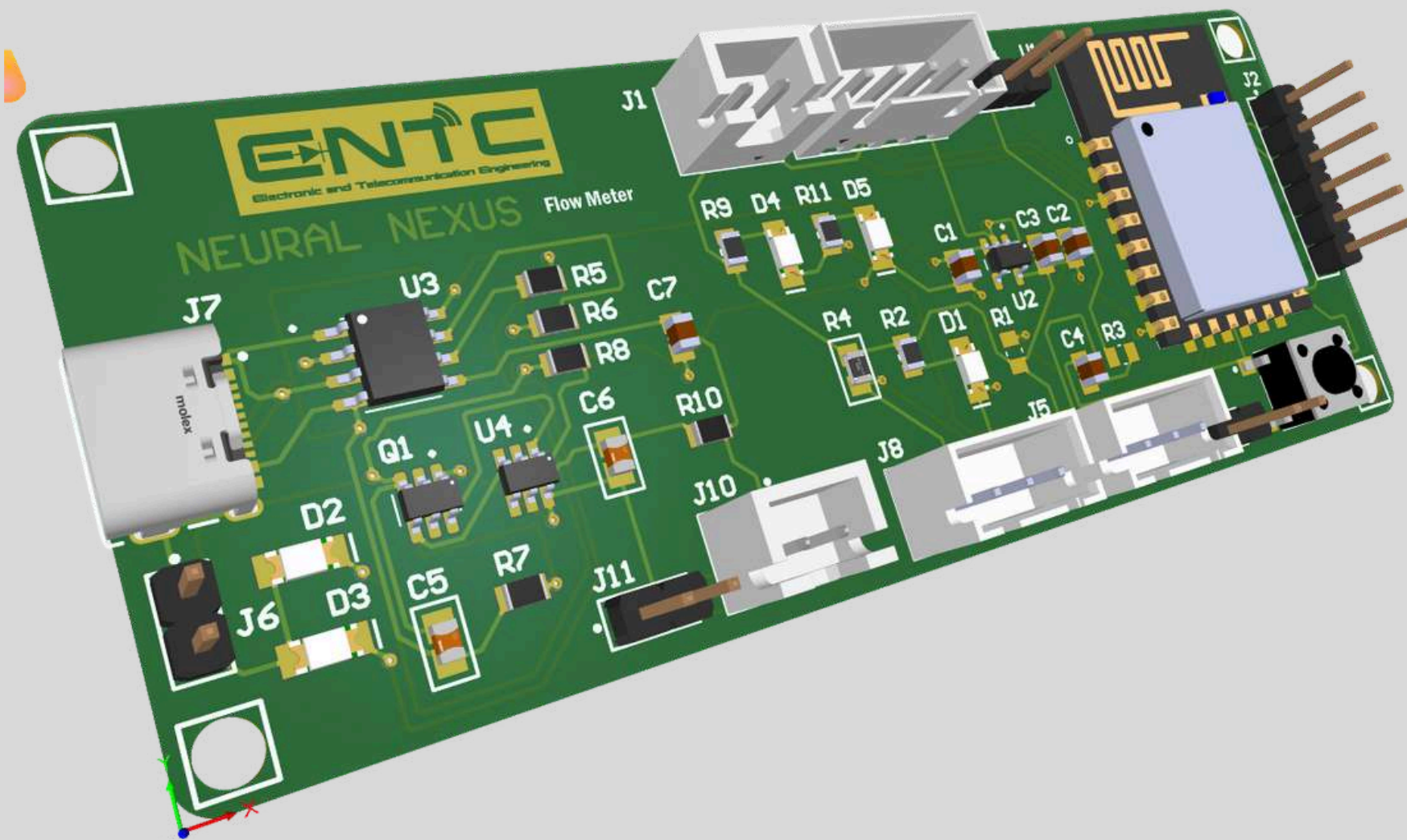
Challenges

High initial investment in advanced flowmeter technology
Complexity of integration in legacy systems
Data security concerns in smart flowmeters
Economic volatility impacting capital expenditure

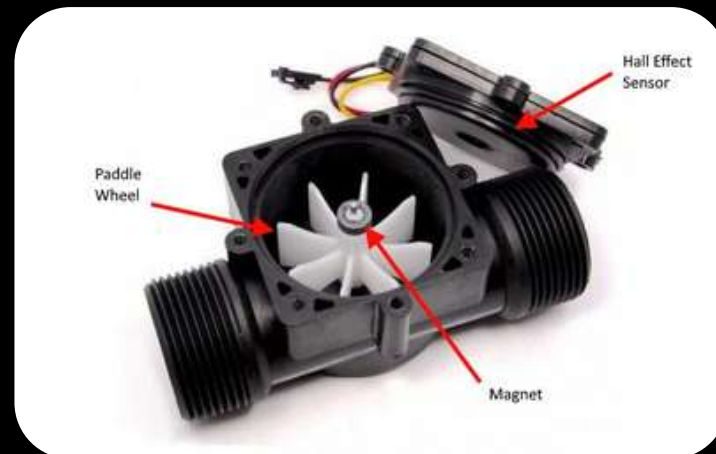
PCB Design



PCB Design

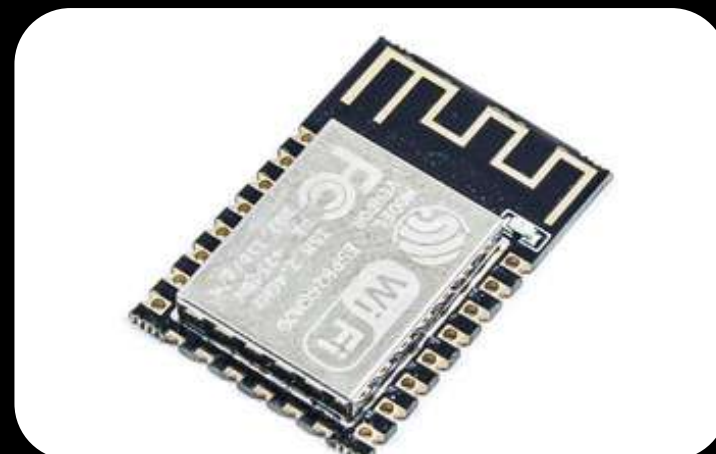


Hardware Components



Paddle wheel Sensor

A paddlewheel sensor measures liquid flow by detecting the rotation of a small wheel placed in the fluid stream. It's simple, cost-effective, and commonly used for water and low-viscosity fluids.



ESP 8266 12 F

The ESP8266-12F is a compact Wi-Fi module with a built-in microcontroller, ideal for IoT applications. It offers reliable wireless connectivity, GPIO pins for sensors, and supports programming via Arduino or Lua.



DS18B20 Temperature Sensor

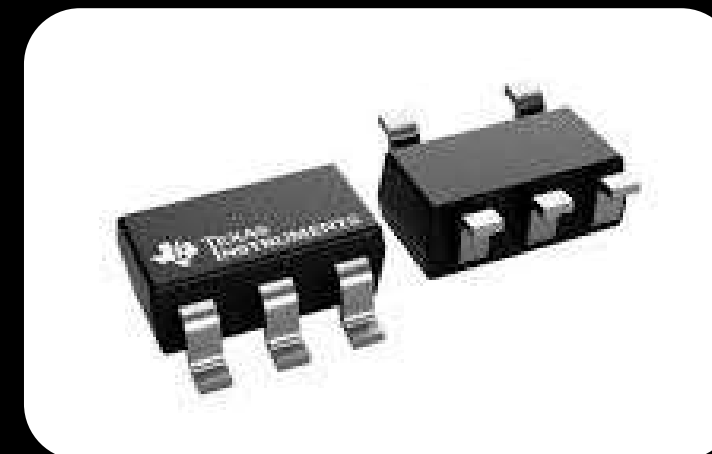
A paddlewheel sensor measures liquid flow by detecting the rotation of a small wheel placed in the fluid stream. It's simple, cost-effective, and commonly used for water and low-viscosity fluids.



Lithium 18650 Batteries

Lithium 18650 Battery

The 18650 is a rechargeable lithium-ion battery, commonly used in electronics, power banks, and EVs. It offers high energy density, long cycle life, and a typical voltage of 3.7V.



TLV75733PDBVR

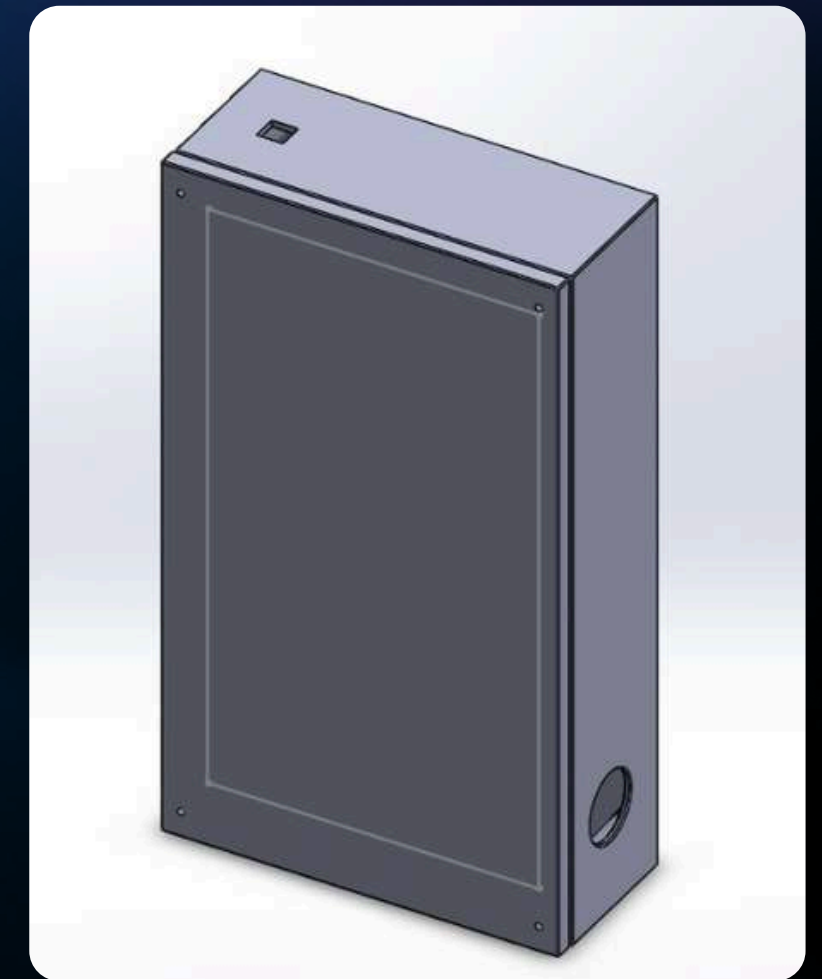
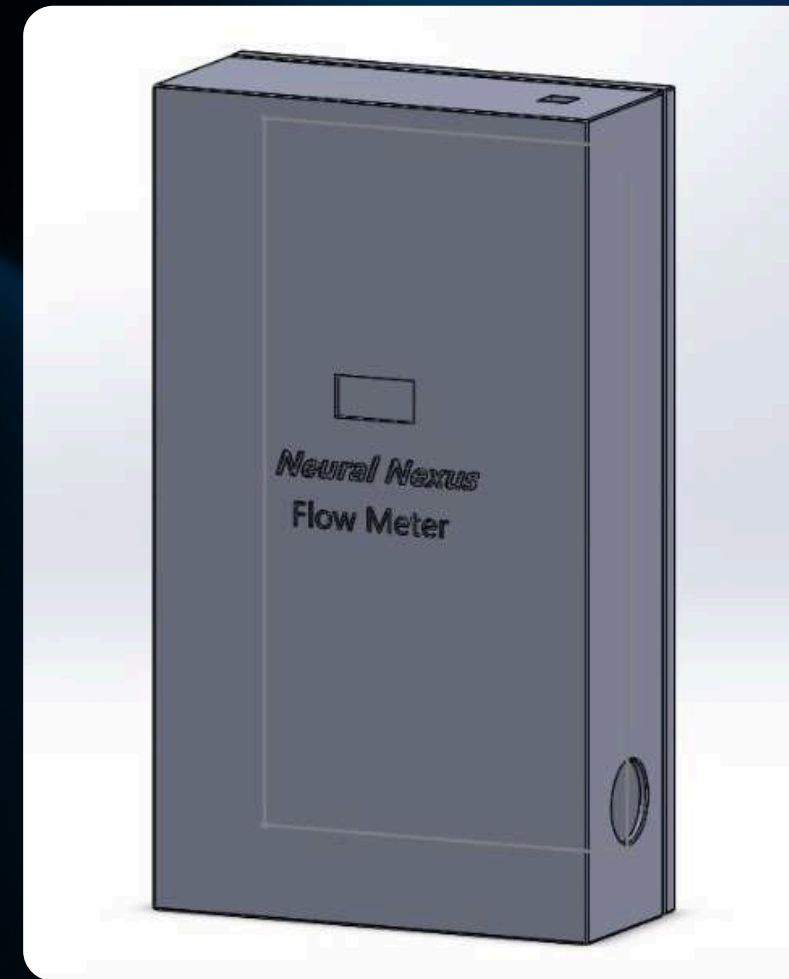
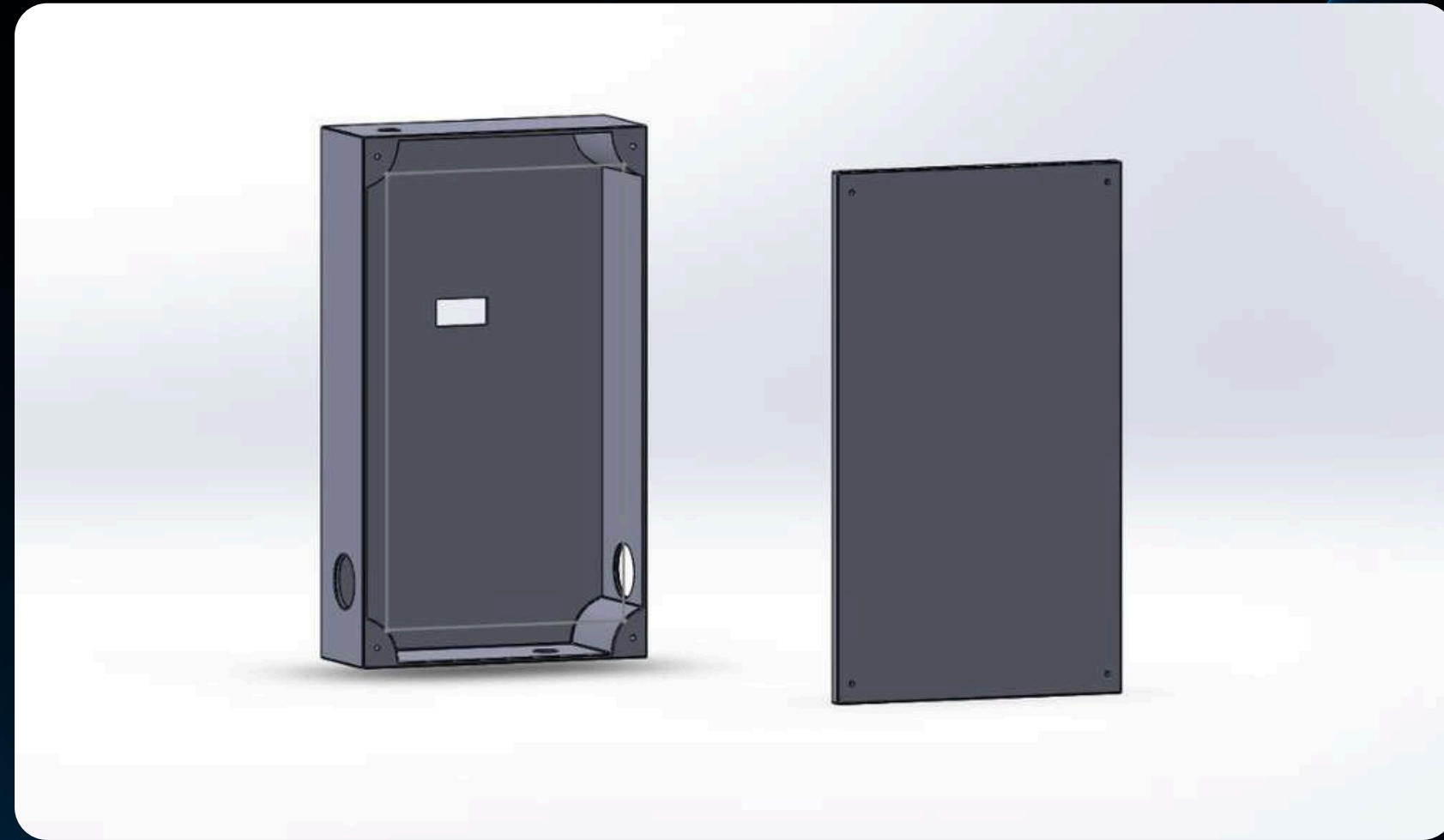
The (LDO) linear voltage regulator from Texas Instruments. It provides a fixed 3.3V output with up to 500 mA current, featuring low quiescent current and thermal protection—ideal for low-power and noise-sensitive applications



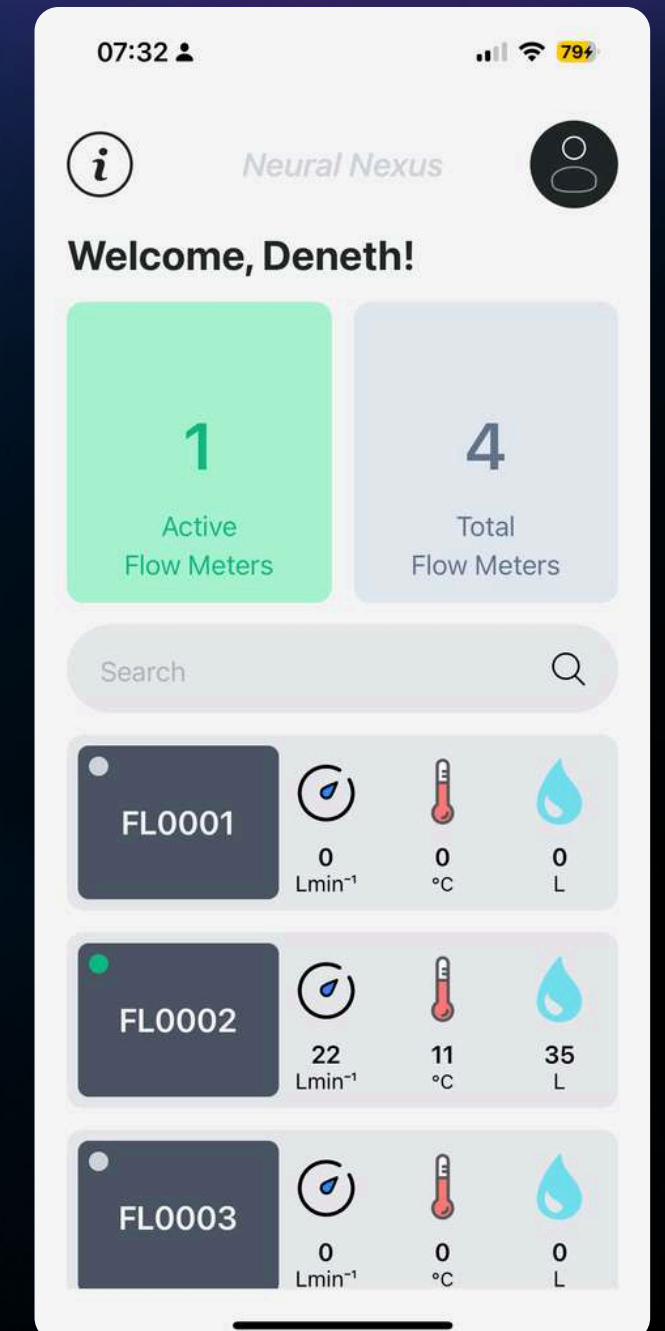
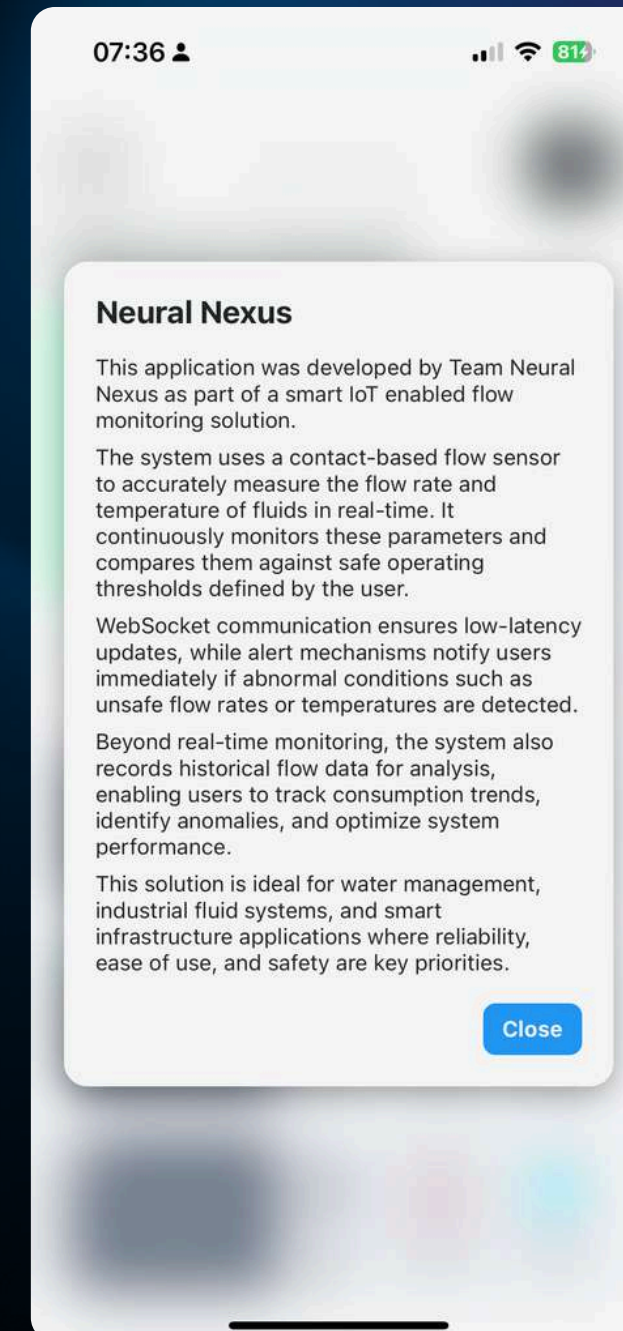
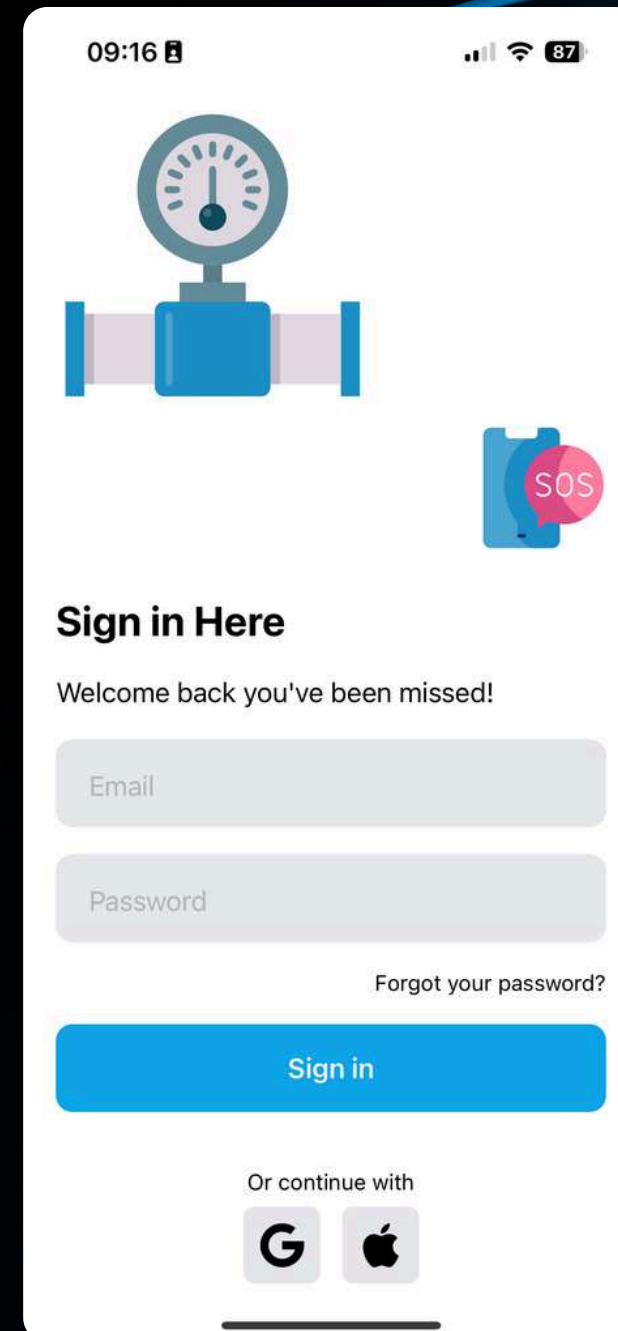
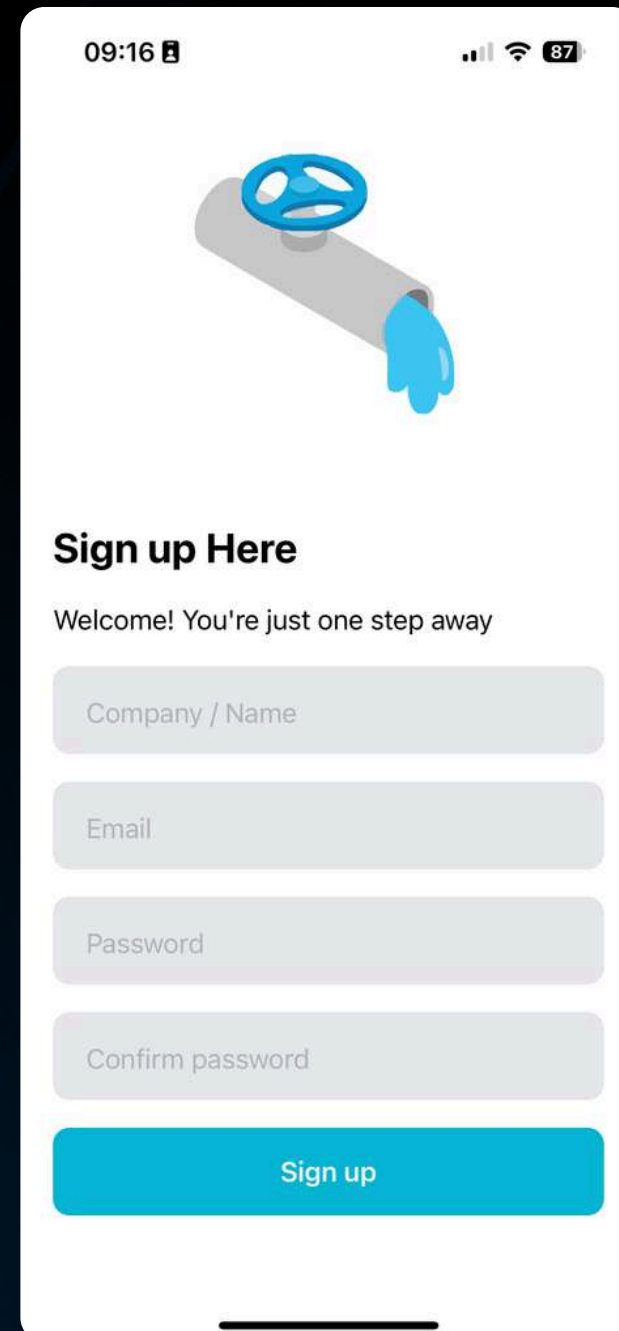
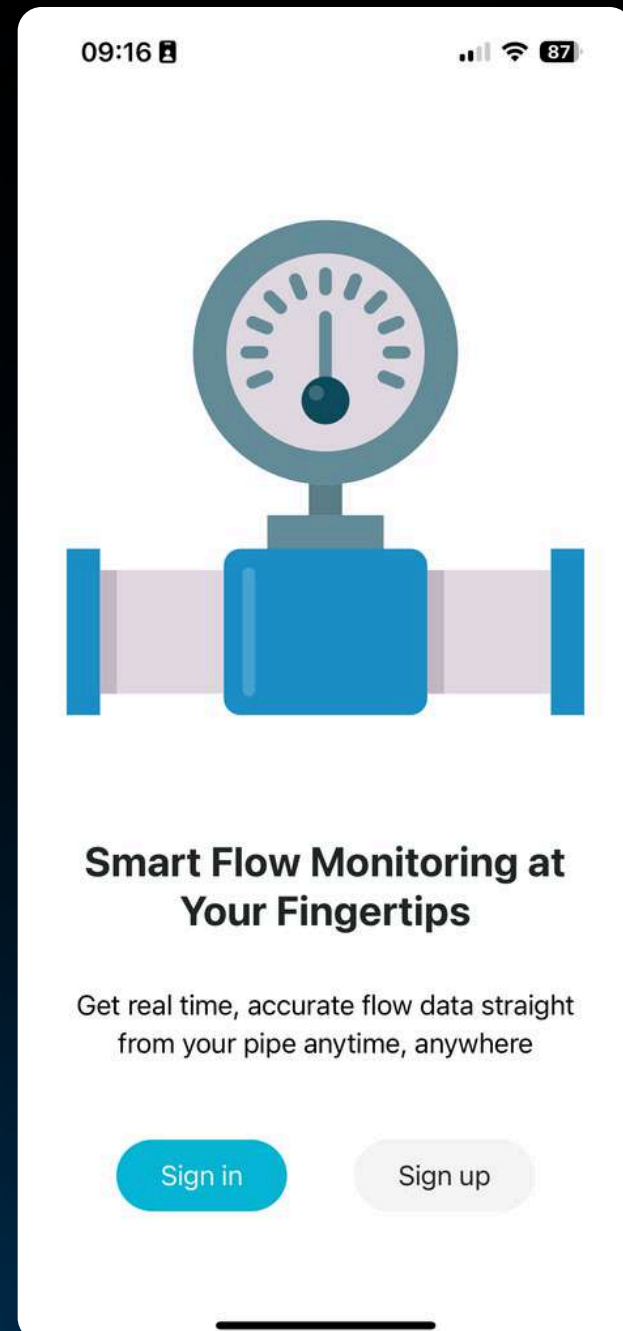
OLED Display

An OLED display is a self-emitting screen that offers high contrast, wide viewing angles, and low power consumption. Common in embedded systems, small OLEDs are often used with microcontrollers for clear, readable output.

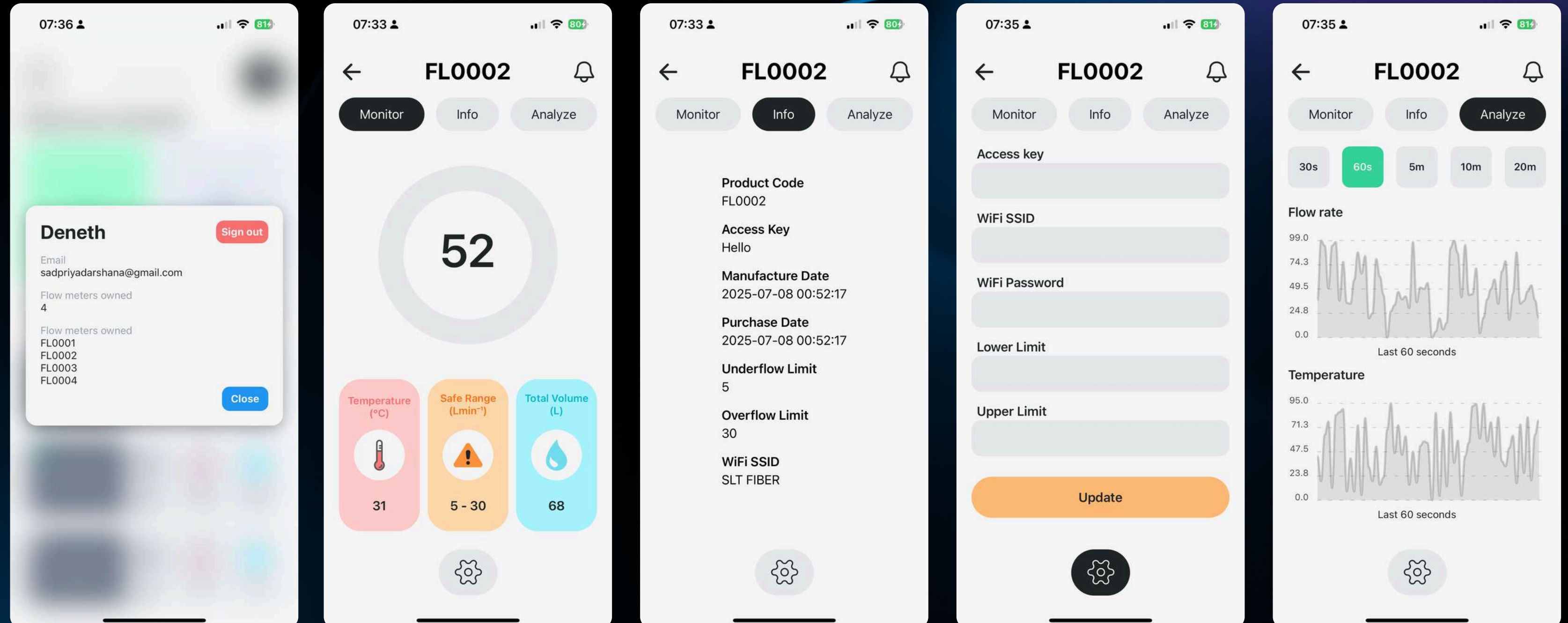
Enclosure Design



App Layout



App Layout



Flowmeter Configuration

Configure Network

SSID

Password

Access Key

Save and Connect